The amazing odd positive integers are those n of the form $n = 3k^2$ (k odd) for which either $3k^2 + 3k + 1$ or $3k^2 - 3k + 1$ is a perfect square. These conditions correspond to Pell equations $x^2 - 3y^2 = 1$ with $x = 2m$, $y = 2k \pm 1$ and yield two infinite families:

- A_t: $n = 3\big((y_{4t-1} - 1)/2\big)^2$ for $t \geq 1$ (from $3k^2 + 3k + 1 = m^2$),
- B_t : $n = 3\big((y_{4t+1} + 1)/2\big)^2$ for $t \geq 1$ (from $3k^2 - 3k + 1 = m^2$), where y_n are the Pell numbers $y_1 = 1$, $y_2 = 4$, $y_3 = 15$, $y_4 = 56$, $y_5 = 209$,... The numbers A_t and B_t satisfy $A_t < B_t < A_{t+1} < B_{t+1}$ for all t, so the increasing list of all amazing odd numbers is $n_1 = A_1 = 147$, $n_2 = B_1 = 33075$, $n_3 = A_2 = 6351075$, $n_4 = B_2 = 1232983587$, $n_5 = A_3 = 239179094643$, $n_6 = B_3 = 46399827249363$, ... Hence the even-indexed terms are $n_{2i} = B_i$ for $i \geq 1$.

From the Pell sequence, $y_{4i+1} = \frac{\alpha^{4i+1} - \beta^{4i+1}}{2\sqrt{3}}$ with $\alpha = 2 + \sqrt{3}$, $\beta = 2 - \sqrt{3}$. Then $n_{2i} = 3\left(\frac{y_{4i+1}+1}{2}\right)^2 = \frac{\alpha^{8i+2}}{16} + \frac{\sqrt{3}}{4}\alpha^{4i+1} + \frac{5}{8} +$ terms that vanish as $i \to \infty$. Thus $n_{2i} \sim \frac{\alpha^2}{16}\,\alpha^{8i}$. Therefore the largest real x for which there exists $y > 0$ with $n_{2i} \geq y\,x^i$ eventually is $x = \alpha^8 = (2 + \sqrt{3})^8$. For this x, the largest z with $n_{2i} \geq z\,x^i$ eventually is $z = \frac{\alpha^2}{16} = \frac{7+4\sqrt{3}}{16}$.

Now compute $x = (2 + \sqrt{3})^8 = 18817 + 10864\sqrt{3}$, $2016z = 2016 \cdot \frac{7+4\sqrt{3}}{16} = 126(7 + 4\sqrt{3}) = 882 + 504\sqrt{3}$, $x + 2016z = (18817 + 882) + (10864 + 504)\sqrt{3} = 19699 + 11368\sqrt{3}$. Thus $p = 19699$, $q = 11368$, $r = 3$ (square-free). Hence $p + q + r = 19699 + 11368 + 3 = 31070$.

$$\boxed{31070}$$